

Small-time Controllability of Control Systems on 3D Lie Groups

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Abstract

We study affine control systems with one control on 3D unimodular Lie groups. Such control systems are first classified into a finite number of equivalent systems. After completing this classification, we determine the small-time controllability of each of the equivalent systems.

1 Introduction

In [1], A. Agrachev considered the Schrödinger equation for the quantum harmonic oscillator, given by

$$\begin{cases} i\partial_t \psi(t, x) &= \left(-\frac{1}{2}\Delta + u_0(t)\frac{|x|^2}{2} + \sum_{j=1}^d u_j(t)x_j \right) \psi(t, x) \\ \psi(t=0) &= \psi_0 \in L^2(\mathbb{R}^d, \mathbb{C}) \end{cases} \quad (1)$$

where the control $u = (u_0, \dots, u_d) \in PWC([0, T], \mathbb{R}^{d+1})$. This equation is highly relevant in quantum dynamics, with applications in physics and chemistry. Based on this equation, the paper studied the controllability of left-invariant control affine systems of the form

$$\frac{d}{dt}q(t) = q(t) \left(a + u_0(t)b + \sum_{i=1}^d u_i(t)X_i + r(t)Z \right) \quad (2)$$

where $q \in SL_2 \ltimes H_d$. Notably, the paper proved that equation (2) is small-time controllable on $SL_2 \ltimes H_d$, and that equation (2) with $u_i, r = 0$ for $i \in \{1, \dots, d\}$ is small-time controllable on SL_2 .

This result motivates the question: does there exist other 3D unimodular Lie groups for which the statement is true? To answer this question, we study affine control systems with one control on 3D unimodular Lie groups, taking the form

$$\dot{q}(t) = (A + u(t)B)q(t) \quad (3)$$

with $q \in G$, $A, B \in \mathfrak{g}$, and u the control. G is a 3D unimodular Lie group, which is known to be completely classified into the Lie groups $SU(2)$, $SL(2)$, H , $SE(2)$, and $SH(2)$. For each, $\mathfrak{g}$ is the corresponding Lie algebra. In section 2, we first classify such systems on each of these Lie groups into equivalent systems. Apparently, there are only finitely many such systems. After this classification is done, we will study controllability. We will begin section 3 by formally defining small-time controllability, among other concepts from geometric control theory. Then, we will characterise the small-time controllability of each of the equivalent systems found. This is a thorough process that will give an answer to the question posed.

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1.1 Convention

Recall the following definitions of the 3D unimodular Lie groups that will be used throughout the paper.

- $SU(2) = \left\{ \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} : a, b \in \mathbb{C}, |a|^2 + |b|^2 = 1 \right\}$
- $SL(2) = \{X \in M_2(\mathbb{R}) : \det X = 1\}$
- $H = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$
- $SE(2) = \left\{ \begin{pmatrix} \cos \theta & -\sin \theta & x \\ \sin \theta & \cos \theta & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, \theta \in \mathbb{R} \right\}$
- $SH(2) = \left\{ \begin{pmatrix} \cosh \theta & \sinh \theta & x \\ \sinh \theta & \cosh \theta & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, \theta \in \mathbb{R} \right\}$

The following are the basis elements chosen for each of the corresponding Lie algebras.

- $\mathfrak{su}(2)$: $\sigma_1 = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$, $\sigma_2 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $\sigma_3 = \frac{1}{2} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$
- $\mathfrak{sl}(2)$: $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
- $\mathfrak{h}$: $X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
- $\mathfrak{se}(2)$: $X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
- $\mathfrak{sh}(2)$: $X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

2 Classifying Control Systems

Consider the control system (3) with $q \in G$ and $A, B \in \mathfrak{g}$, as formulated in the introduction. When classifying (3) on a particular Lie group, we want to simplify the pair (A, B) into an equivalent pair, say (f_0, f_1) . This equivalency is denoted $(A, B) \sim (f_0, f_1)$.

What tools can we use in this classification? The first is the adjoint action of a matrix Lie group on its Lie algebra. Letting g act on both A and B , the Killing form $B(A, B)$ is invariant

$$B(gAg^{-1}, gBg^{-1}) = B(A, B)$$

Thus $(A, B) \sim (gAg^{-1}, gBg^{-1})$. Second, the control can be shifted by δ , so

$$A + uB = A + (u' + \delta)B = (A + \delta B) + u'B$$

The control can also be scaled by a constant μ , so

$$A + uB = A + (\mu u')B = A + u'(\mu B)$$

Finally, the time can be scaled by a factor of $1/\lambda$. The equivalences they imply are summarised in the following lemma.

Lemma 2.1. For the control system (3), we have the following equivalence relations:

1. $(A, B) \sim (gAg^{-1}, gBg^{-1})$
2. $(A, B) \sim ((A + \delta B), B)$
3. $(A, B) \sim (A, \mu B)$
4. $(A, B) \sim (\lambda A, \lambda B)$

Similar work regarding the classification of control systems may have already been partially done by C. Remsing, for example in [3], [4], [5]. They notably worked on the $SU(2)$, $SL(2)$, H , and $SE(2)$ groups, but used Lie group isomorphisms and Lie algebra isomorphisms, unlike what we are doing in this paper. This may cause the results to differ.

Remark. We will always assume that B is nonzero, and that A and B are independent.

2.1 Classifying Control Systems on $SU(2)$

We consider the system (3) with $q \in SU(2)$ and $A, B \in \mathfrak{su}(2)$. This is the most basic scenario as it is closely related to the usual Euclidean space $\mathbb{R}^3$.

Let $B(\cdot, \cdot)$ denote the Killing form. For $A, B \in \mathfrak{su}(2)$, $B(A, A)$ and $B(B, B)$ represent the lengths of A and B respectively, and $B(A, B)$ represents the angle between A and B (all in the sense of the usual Euclidean geometry). One can check that these quantities are preserved under conjugation, suggesting that the conjugation group action acts as rotations in $\mathbb{R}^3$.

Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 \sigma_1 + a_2^0 \sigma_2 + a_3^0 \sigma_3, \quad B = b_1^0 \sigma_1 + b_2^0 \sigma_2 + b_3^0 \sigma_3.$$

Construct a 3D space with $\sigma_1, \sigma_2, \sigma_3$ as the axes, analogous to $\mathbb{R}^3$. Applying a combination of conjugation group actions to both A and B , we can rotate B onto the σ_3 axis, then rotate A onto the σ_1 - σ_3 plane.

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 \sigma_1 + a_3^1 \sigma_3, b_3^1 \sigma_3) \\ &\sim (a_1^1 \sigma_1, b_3^1 \sigma_3) \\ &\sim (a_1^1 \sigma_1, a_1^1 \sigma_3) \\ &\sim (\sigma_1, \sigma_3) \end{aligned}$$

Hence, all control systems (3) on $SU(2)$ are equivalent to (σ_1, σ_3) .

2.2 Classifying Control Systems on $SL(2)$

We consider the system (3) with $g \in SL(2, \mathbb{R})$ and $A, B \in \mathfrak{sl}(2)$. Recall that for matrices $u_i, u_j \in \mathfrak{sl}(2)$, the matrix of the Killing form is the matrix K whose entries are $k_{ij} = B(u_i, u_j)$. Using the aforementioned basis elements, we compute the Killing form matrix to be

$$K = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 8 \end{pmatrix}$$

The eigenvalues of this matrix are -4, 4, and 8. There are 2 positive eigenvalues and 1 negative eigenvalue, so the Killing form has signature (2, 1). From the metric signature (2, 1), we see that $SL(2)$ is a group consisting of transformations in Lorentzian space $\mathbb{R}^{2,1}$.

In Lorentzian space, vectors can be classified as elliptic, hyperbolic, or parabolic based on the sign of their norm, or equivalently, the sign of the Killing form (analogous to the Minkowski scalar product). Let $X \in \mathfrak{sl}(2)$. If $B(X, X) = 0$, then X is parabolic, and lies on the light cone. If $B(X, X) < 0$, then X is elliptic, and lies on the hyperboloid of two sheets inside the light cone. If $B(X, X) > 0$, then X is hyperbolic, and lies on the hyperboloid of one sheet outside the light cone.

Lorentz transformations - including null rotations, rotations, and Lorentz boosts - act via the conjugation group action. There does not exist a Lorentz transformation from the upper or lower sheet of the hyperboloid of two sheets to the hyperboloid of one sheet. There also does not exist a Lorentz transformation the upper to the lower sheet of the hyperboloid of two sheets. In short, a vector can only remain in the region of space where it started.

The strategy for classification is to begin by using the sign of the Killing form of B to split the analysis into three major cases. In each of these cases, we can identify a representative based on the sign of the Killing form. We choose E , H , $E - F$ as the representatives for the parabolic, hyperbolic, and elliptic cases, respectively. Since this representative and B lie in the same region of space, we can apply Lorentz transformations via conjugation to simplify B to this representative.

Remark. If, in our convention, E is the representative for the upper half of the light cone, then $-E$ is the representative for the lower half of the light cone. The same goes for the hyperboloid of two sheets, by taking $E - F$ and $F - E$. However, it is not necessary to discuss the upper and lower halves separately. One can be transformed to the other by simply scaling the control.

Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 E + a_2^0 F + a_3^0 H, \quad B = b_1^0 E + b_2^0 F + b_3^0 H.$$

Case 1: $B(B, B) > 0$, B is hyperbolic

By a Lorentz transformation, B can be simplified to a multiple of H .

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 = 0$, then

$$\begin{aligned} (A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \\ &= (a_1^1 E + a_3^1 H, b_3^1 H) \\ &\sim (a_1^1 E, b_3^1 H) \\ &\sim (a_1^1 E, a_1^1 H) \\ &\sim (E, H) \end{aligned}$$

If $a_1^1 = 0$ and $a_2^1 \neq 0$, then

$$\begin{aligned} (A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \\ &= (a_2^1 F + a_3^1 H, b_3^1 H) \\ &\sim (a_2^1 F, b_3^1 H) \\ &\sim (a_2^1 F, a_2^1 H) \\ &\sim (F, H) \end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 \neq 0$, then we need to find elements in the stabiliser group of H to continue simplifying while preserving H . Such elements of $Stab(H)$ must be of the form

$$g = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}, \quad ad = 1$$

which represents a Lorentz boost. Applying to $(a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H)$, we have

$$\begin{aligned} (A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \\ &\sim (a_1^1 a^2 E + a_2^1 d^2 F + a_3^1 H, b_3^1 H) \\ &\sim (c(E + F), b_3^1 H) \\ &\sim (c(E + F), cH) \\ &\sim (E + F, H) \end{aligned}$$

where we chose a and d such that $a_1^1 a^2 = a_2^1 d^2 = c$. The precise choices for a and d can be found by using $a_1^1 a^2 = a_2^1 d^2$ along with $ad = 1$.

Taking note of the sign, there is also $(-E - F, H)$, but this can be transformed into $(E + F, H)$ be an outer automorphism. Precisely, we have

$$Aut(\mathfrak{sl}(2)) \cong PGL(2, \mathbb{R})$$

and we can take

$$D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and perform the mapping $g \mapsto DgD^{-1}$ for $g \in \mathfrak{sl}(2)$.

Case 2: $B(B, B) < 0$, B is elliptic

By a Lorentz transformation, B can be simplified to a multiple of $E - F$.

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 E + a_2^1 F + a_3^1 H, b_1^1 (E - F)) \end{aligned}$$

To continue simplifying, we want to find elements of the stabiliser group $Stab(E - F)$. These elements take the form

$$g = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}, \quad a^2 + b^2 = 1$$

which resembles a rotation matrix, and so the Lorentz transformations in this case act as rotations in a plane. If we fix $E - F$ as one axis, then for a 3D space, we need to construct two more, and it is on the plane formed by these two axes that we will rotate. Consider

$$E + F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

By computing the Killing forms, one can check that $E - F$, $E + F$, and H are indeed orthogonal, forming a set of coordinate axes. Finding these two remaining axes can be thought of as finding the orthogonal complement of B ,

$$B^\perp = \{X \in \mathfrak{sl}(2) : B(B, X) = 0\} = \text{span}\{E + F, H\}$$

The strategy is then to rewrite A and B in terms of this new set of coordinates, reduce A onto $B^\perp$ by control shifting, then rotating on $B^\perp$. The steps are as follows:

$$\begin{aligned}
(A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_1^1 (E - F)) \\
&\sim (a_1^2 (E + F) + a_3^2 H, b_1^1 (E - F)) \\
&\sim (a_1^3 (E + F), b_1^1 (E - F)) \\
&\sim (a_1^3 (E + F), a_1^3 (E - F)) \\
&\sim (2a_1^3 E, a_1^3 (E - F)) \\
&\sim (2a_1^3 E, 2a_1^3 (E - F)) \\
&\sim (E, E - F)
\end{aligned}$$

Remark. We could have equivalently rotated to H , which would give $(H, E - F)$. It does not matter which one we rotate to. Indeed, we always have $B(A, A) > 0$, hence rotations are all occurring on the hyperboloid of one sheet.

Case 3: $B(B, B) = 0$, B is parabolic

By a Lorentz transformation, B can be reduced to a multiple of E .

$$\begin{aligned}
(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\
&= (a_1^1 E + a_2^1 F + a_3^1 H, b_1^1 E)
\end{aligned}$$

Again, we find elements of the stabiliser group of E , $Stab(E)$. Such elements are of the form

$$g = \begin{pmatrix} a & b \\ 0 & a \end{pmatrix}, \quad a^2 = 1$$

resembling the null rotation. Applying this to $(a_1^1 E + a_2^1 F + a_3^1 H, b_1^1 E)$,

$$(A, B) \sim ((a_1^1 - b^2 a_2^1 - 2aba_3^1)E + a_2^1 F + (aba_2^1 + a_3^1)H, b_1^1 E)$$

If $a_2^1 = 0$, then

$$\begin{aligned}
(A, B) &\sim (a_3^1 H, b_1^1 E) \\
&\sim (a_3^1 H, a_3^1 E) \\
&\sim (H, E)
\end{aligned}$$

If $a_2^1 \neq 0$, then choose $b = -a_3^1/aa_2^1$ to eliminate the H component in A , giving

$$\begin{aligned}
(A, B) &\sim (a_2^1 F, b_1^1 E) \\
&\sim (a_2^1 F, a_2^1 E) \\
&\sim (F, E)
\end{aligned}$$

2.3 Classifying Control Systems on H

We consider the system (3) with $q \in H$ and $A, B \in \mathfrak{h}$. We start by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z, \quad B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

The Heisenberg group is different from the previous two scenarios in that for any $B \in \mathfrak{h}$, the Killing form is degenerate. We use other transformations instead. The first are inner

automorphisms. These preserve the X and Y components, only shifting the Z component. Applying this transformation to A and B , we obtain

$$gAg^{-1} = a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, \quad gBg^{-1} = b_1^0 X + b_2^0 Y + (-bb_1^0 + ab_2^0 + b_3^0)Z$$

The second type are symplectic maps. The symplectic map uses a matrix $T \in Sp(2, \mathbb{R})$ to act by left matrix multiplication on the (X, Y) vector in M , and preserves the Z component. Symbolically, we can write the mapping

$$((X, Y), Z) \mapsto (T(X, Y), Z)$$

Note that in the $n = 2$ case, $Sp(2, \mathbb{R}) = SL(2, \mathbb{R})$. So when $T \in Sp(2, \mathbb{R})$, we can take T to be the rotation matrix

$$T = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Doing so allows us to rotate in the (X, Y) plane.

Consider also the centre of the Lie algebra $\mathfrak{h}$, which is

$$\mathfrak{z}(\mathfrak{h}) = \left\{ \begin{pmatrix} 0 & 0 & c \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} : c \in \mathbb{R} \right\} = \{cZ : c \in \mathbb{R}\}$$

We try to simplify B first, and to do so, we analyze two cases: either B belongs to the centre, or B does not belong to the centre.

Case 1: $B \in \mathfrak{z}(\mathfrak{h})$

Here, $b_3^0 \neq 0$, while $b_1^0 X + b_2^0 Y = 0$, i.e. $b_1^0 = b_2^0 = 0$. Then,

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_3^0 Z) \\ &\sim (a_1^0 X + a_2^0 Y, b_3^0 Z) \end{aligned}$$

It must be that $(a_1^0, a_2^0) \neq (0, 0)$. Using the rotation matrix T simultaneously on A and B , we can rotate $a_1^0 X + a_2^0 Y$ onto either X or Y , while $b_3^0 Z$ remains unchanged.

$$\begin{aligned} (A, B) &\sim (a_1^1 X, b_3^0 Z) \\ &\sim (a_1^1 X, a_1^1 Z) \\ &\sim (X, Z) \end{aligned}$$

Case 2: $B \notin \mathfrak{z}(\mathfrak{h})$

Here, $b_1^0 X + b_2^0 Y \neq 0$, so $(b_1^0, b_2^0) \neq (0, 0)$. b_3^0 may or may not be 0. In the inner automorphism on B ,

$$gBg^{-1} = b_1^0 X + b_2^0 Y + (-bb_1^0 + ab_2^0 + b_3^0)Z$$

Since $(b_1^0, b_2^0) \neq (0, 0)$, we can choose the entries $a, b \in \mathbb{R}$ of the matrix g such that $-bb_1^0 + ab_2^0 + b_3^0 = 0$, and so

$$gBg^{-1} = b_1^0 X + b_2^0 Y$$

This can further be rotated on to X by applying the symplectic map to A and B . That is,

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \\ &\sim (a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \end{aligned}$$

If $a_2^1 = 0$, then

$$\begin{aligned}(A, B) &= ((-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1X) \\ &= ((-ba_1^0 + aa_2^0 + a_3^0)Z, (-ba_1^0 + aa_2^0 + a_3^0)X) \\ &= (Z, X)\end{aligned}$$

If $a_2^1 \neq 0$, then we need another inner automorphism that is at the same time an element of the stabiliser of X . This is $h \in H$ of the form

$$h = \begin{pmatrix} 1 & a' & c' \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \text{Stab}(X)$$

We now apply this inner automorphism.

$$\begin{aligned}(A, B) &\sim (a_2^1Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1X) \\ &\sim (h(a_2^1Y + (-ba_1^0 + aa_2^0 + a_3^0)Z)h^{-1}, h(b_1^1X)h^{-1}) \\ &= (a_2^1Y + (a_2^1a' + (-ba_1^0 + aa_2^0 + a_3^0))Z, b_1^1X)\end{aligned}$$

Since $a_2^1 \neq 0$, we can choose the entries of h to be $a' = -(-ba_1^0 + aa_2^0 + a_3^0)/a_2^1$, $c' \in \mathbb{R}$ so that $a_2^1a' + (-ba_1^0 + aa_2^0 + a_3^0) = 0$. Proceeding with this,

$$\begin{aligned}(A, B) &\sim (a_2^1Y, b_1^1X) \\ &\sim (a_2^1Y, a_2^1X) \\ &\sim (Y, X)\end{aligned}$$

2.4 Classifying Control Systems on $SE(2)$

We consider the system (3) with $q \in SE(2)$ and $A, B \in \mathfrak{se}(2)$. Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0X + a_2^0Y + a_3^0Z, \quad B = b_1^0X + b_2^0Y + b_3^0Z$$

For a matrix $g \in SE(2)$, the conjugation group action on the basis elements gives

$$gXg^{-1} = \cos \theta X + \sin \theta Y, \quad gYg^{-1} = -\sin \theta X + \cos \theta Y, \quad gZg^{-1} = yX - xY + Z$$

with $x, y, \theta \in \mathbb{R}$. Indeed, the transformations are combinations of rotations and translations in Euclidean space, with rotations determined by θ , and translations determined by x and y .

On A and B , we have

$$\begin{aligned}gAg^{-1} &= (a_1^0 \cos \theta - a_2^0 \sin \theta + a_3^0 y)X + (a_1^0 \sin \theta + a_2^0 \cos \theta - a_3^0 x)Y + a_3^0 Z \\ gBg^{-1} &= (b_1^0 \cos \theta - b_2^0 \sin \theta + b_3^0 y)X + (b_1^0 \sin \theta + b_2^0 \cos \theta - b_3^0 x)Y + b_3^0 Z\end{aligned}$$

There are two cases to consider: when $B \in \text{Span}(X, Y)$, and when $B \notin \text{Span}(X, Y)$.

Case 1: $B \in \text{Span}(X, Y)$

In this case, $B = b_1^0X + b_2^0Y$, with $(b_1^0, b_2^0) \neq 0$ and $b_3^0 = 0$. Since b_3^0 , we can disregard x and y , and choose θ so that gBg^{-1} rotates B in the (X, Y) plane on to X , then simplify.

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= ((a_1^0 \cos \theta - a_2^0 \sin \theta + a_3^0 y)X + (a_1^0 \sin \theta + a_2^0 \cos \theta - a_3^0 x)Y + a_3^0 Z, b_1^1X) \\ &= (a_1^1X + a_2^1Y + a_3^0Z, b_1^1X) \\ &\sim (a_2^1Y + a_3^0Z, b_1^1X)\end{aligned}$$

If $a_3^0 \neq 0$, then we need to find the elements of $Stab(X)$, so that conjugating preserves $b_1^1 X$. It is clear that we require $\theta = 0$. Moreover, noting the result of gZg^{-1} , we choose $x = a_3^1/a_3^0$ and $y = 0$ to eliminate the Y component and prevent X from reappearing in A . Thus,

$$\begin{aligned}(A, B) &\sim (a_3^0 Z, b_1^1 X) \\ &\sim (a_3^0 Z, a_3^0 X) \\ &\sim (Z, X)\end{aligned}$$

If $a_3^0 = 0$, then assuming $a_2^1 \neq 0$ (otherwise we obtain the same result as before),

$$\begin{aligned}(A, B) &\sim (a_2^1 Y, b_1^1 X) \\ &\sim (a_2^1 Y, a_2^1 X) \\ &\sim (Y, X)\end{aligned}$$

Case 2: $B \notin Span(X, Y)$

Here, $b_3^0 \neq 0$. When applying inner automorphisms, we first choose $x = 0$ and $y = 0$ so that we only rotate in the X - Y plane.

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= ((a_1^0 \cos \theta - a_2^0 \sin \theta + a_3^0 y)X + (a_1^0 \sin \theta + a_2^0 \cos \theta - a_3^0 x)Y + a_3^0 Z, b_1^1 X + b_3^0 Z) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X + b_3^0 Z)\end{aligned}$$

If $b_1^1 = 0$, then

$$\begin{aligned}(A, B) &\sim (a_1^1 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^1 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

Apply a second inner automorphism, choosing $x = 0$, $y = 0$, and $\theta \in \mathbb{R}$ to rotate onto X .

$$\begin{aligned}(A, B) &\sim (a_1^2 X, b_3^0 Z) \\ &\sim (a_1^2 X, a_1^2 Z) \\ &\sim (X, Z)\end{aligned}$$

If $b_1^1 \neq 0$, we take another $g \in SE(2)$ with $\theta = 0$ so that the inner automorphism causes no rotation, only translation. Choosing $x = 0$, $y = -b_1^1/b_3^0$,

$$\begin{aligned}(A, B) &\sim (g(a_1^1 X + a_2^1 Y + a_3^0 Z)g^{-1}, g(b_1^1 X + b_3^0 Z)g^{-1}) \\ &= ((a_1^1 + a_3^0 y)X + (a_2^1 - a_3^0 x)Y + a_3^0 Z, (b_1^1 + b_3^0 y)X - b_3^0 xY + b_3^0 Z) \\ &\sim (a_1^2 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^2 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

Finally, use one last inner automorphism and set $x = 0$, $y = 0$. $\theta \in \mathbb{R}$ is chosen such that $a_1^2 X + a_2^1 Y$ is rotated on to X .

$$\begin{aligned}(A, B) &\sim (a_1^3 X, b_3^0 Z) \\ &\sim (a_1^3 X, a_1^3 Z) \\ &\sim (X, Z)\end{aligned}$$

2.5 Classifying Control Systems on $SH(2)$

Consider the system (3) with $q \in SH(2)$ and $A, B \in \mathfrak{sh}(2)$. The $SH(2)$ group is similar to $SE(2)$, except that $SH(2)$ is translations and hyperbolic rotations preserving the Lorentzian.

Again, begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z, \quad B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

For a matrix $g \in SH(2)$, the conjugation group action on the basis elements gives

$$gXg^{-1} = \cosh \theta X + \sinh \theta Y, \quad gYg^{-1} = \sinh \theta X + \cosh \theta Y, \quad gZg^{-1} = -yX - xY + Z$$

with $x, y, \theta \in \mathbb{R}$. Indeed, these transformations are combinations of hyperbolic rotations determined by θ , and translations determined by x and y .

On A and B , we have

$$gAg^{-1} = (a_1^0 \cosh \theta + a_2^0 \sinh \theta - a_3^0 y)X + (a_1^0 \sinh \theta + a_2^0 \cosh \theta - a_3^0 x)Y + a_3^0 Z$$

$$gBg^{-1} = (b_1^0 \cosh \theta + b_2^0 \sinh \theta - b_3^0 y)X + (b_1^0 \sinh \theta + b_2^0 \cosh \theta - b_3^0 x)Y + b_3^0 Z$$

The analysis requires two cases to be considered: $B \in \text{Span}(X, Y)$ and $B \notin \text{Span}(X, Y)$.

Case 1: $B \in \text{Span}(X, Y)$

Here, $b_3^0 = 0$ and $(b_1^0, b_2^0) \neq (0, 0)$. If $b_1^0 \neq 0$ and $b_2^0 = 0$, then

$$\begin{aligned} (A, B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_1^0 X) \\ &\sim (a_2^0 Y + a_3^0 Z, b_1^0 X) \end{aligned}$$

Now, if $a_3^0 = 0$,

$$\begin{aligned} (A, B) &\sim (a_2^0 Y, b_1^0 X) \\ &\sim (a_2^0 Y, a_2^0 X) \\ &\sim (Y, X) \end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, we apply a transformation consisting only of a shift in Y . This is done by choosing $\theta = 0$, $y = 0$, and $x = a_2^0/a_3^0$.

$$\begin{aligned} (A, B) &\sim (g(a_2^0 Y + a_3^0 Z)g^{-1}, g(b_1^0 X)g^{-1}) \\ &= ((a_2^0 - a_3^0 x)Y + a_3^0 Z, b_1^0 X) \\ &= (a_3^0 Z, b_1^0 X) \\ &\sim (a_3^0 Z, a_3^0 X) \\ &\sim (Z, X) \end{aligned}$$

If $b_1^0 = 0$ and $b_2^0 \neq 0$, then

$$\begin{aligned} (A, B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_2^0 Y) \\ &\sim (a_1^0 X + a_3^0 Z, b_2^0 Y) \end{aligned}$$

If $a_3^0 = 0$, then

$$\begin{aligned} (A, B) &\sim (a_1^0 X, b_2^0 Y) \\ &\sim (a_1^0 X, a_1^0 Y) \\ &\sim (X, Y) \end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, we apply a transformation consisting only of a shift in X . This is done by choosing $\theta = 0$, $x = 0$, and $y = a_1^0/a_3^0$.

$$\begin{aligned}(A, B) &\sim (g(a_1^0 X + a_3^0 Z)g^{-1}, g(b_2^0 Y)g^{-1}) \\ &= ((a_1^0 - a_3^0 y)X + a_3^0 Z, b_2^0 Y) \\ &= (a_3^0 Z, b_2^0 Y) \\ &\sim (a_3^0 Z, a_3^0 Y) \\ &\sim (Z, Y)\end{aligned}$$

The most general case is $b_1^0 \neq 0$ and $b_2^0 \neq 0$. We try to use hyperbolic rotations to simplify B . Referring to the form of B after a hyperbolic rotation,

$$gBg^{-1} = (b_1^0 \cosh \theta + b_2^0 \sinh \theta)X + (b_1^0 \sinh \theta + b_2^0 \cosh \theta)Y,$$

if we want to reduce one of the coefficients to 0, we need to solve for θ in $\tanh \theta = -b_1^0/b_2^0$ or $-b_2^0/b_1^0$. $\tanh$ is a mapping $\mathbb{R} \mapsto (-1, 1)$. There is no solution when $b_1^0 = b_2^0$. So three different cases need to be analyzed, depending on the ratio of b_1^0 and b_2^0 .

If $b_1^0 = b_2^0$, then

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^0 X + b_1^0 Y)\end{aligned}$$

Now if $a_3^0 = 0$, then

$$\begin{aligned}(A, B) &\sim (a_1^1 X + a_2^1 Y, b_1^0 X + b_1^0 Y) \\ &\sim (a_1^2 X, b_1^0(X + Y)) \\ &\sim (a_1^2 X, a_1^2(X + Y)) \\ &\sim (X, X + Y)\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, take another $g \in SH(2)$ with $\theta = 0$ so that there is no rotation. Applying the transformation gives

$$(A, B) \sim ((a_1^1 - a_3^0 y)X + (a_2^1 - a_3^0 x)Y + a_3^0 Z, b_1^0(X + Y))$$

Since $a_3^0 \neq 0$, it is possible to choose $x, y \in \mathbb{R}$ such that $a_1^1 - a_3^0 y = 0$ and $a_2^1 - a_3^0 x = 0$. Hence

$$\begin{aligned}(A, B) &\sim (a_3^0 Z, b_1^0(X + Y)) \\ &\sim (a_3^0 Z, a_3^0(X + Y)) \\ &\sim (Z, X + Y)\end{aligned}$$

If $b_1^0 < b_2^0$, then the equation $b_1^0 \cosh \theta + b_2^0 \sinh \theta = 0$ has a solution. Proceeding with such as θ , we obtain (X, Y) and (Z, Y) . If $b_2^0 < b_1^0$, then the equation $b_1^0 \sinh \theta + b_2^0 \cosh \theta = 0$ has a solution. Proceeding with such as θ , we obtain (Y, X) and (Z, X) . These equivalent systems have already been encountered.

Case 2: $B \notin \text{Span}(X, Y)$

Here, $b_3^0 \neq 0$. Recalling the general form of gAg^{-1} and gBg^{-1} that was initially derived, it is possible to choose $x, y \in \mathbb{R}$ such that the coefficients $b_1^0 \cosh \theta + b_2^0 \sinh \theta - b_3^0 y = 0$ and $b_1^0 \sinh \theta + b_2^0 \cosh \theta - b_3^0 x = 0$, causing the X and Y components in B to vanish. We have

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^1 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 = 0$, then

$$\begin{aligned}(A, B) &\sim (a_1^1 X, b_3^0 Z) \\ &\sim (a_1^1 X, a_1^1 Z) \\ &\sim (X, Z)\end{aligned}$$

If $a_1^1 = 0$ and $a_2^1 \neq 0$, then

$$\begin{aligned}(A, B) &\sim (a_2^1 Y, b_3^0 Z) \\ &\sim (a_2^1 Y, a_2^1 Z) \\ &\sim (Y, Z)\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 \neq 0$, then just as before, there are three cases to analyse. If $a_1^1 = a_2^1$, then

$$\begin{aligned}(A, B) &\sim (a_1^1(X + Y), b_3^0 Z) \\ &\sim (a_1^1(X + Y), a_1^1 Z) \\ &\sim (X + Y, Z)\end{aligned}$$

Otherwise, in order to simplify further, we require an element of the stabiliser group $Stab(Z)$. It is clear that for $g \in SH(2)$, we must take $x = 0$ and $y = 0$. We have

$$\begin{aligned}(A, B) &\sim (g(a_1^1 X + a_2^1 Y)g^{-1}, g(b_3^0 Z)g^{-1}) \\ &= ((a_1^1 \cosh \theta + a_2^1 \sinh \theta)X + (a_1^1 \sinh \theta + a_2^1 \cosh \theta)Y, b_3^0 Z)\end{aligned}$$

We follow the same reasoning as before and try to apply rotations to simplify. If $a_1^1 < a_2^1$, then there exists $\theta \in \mathbb{R}$ such that $a_1^1 \cosh \theta + a_2^1 \sinh \theta = 0$. Hence

$$\begin{aligned}(A, B) &\sim (a_2^2 Y, b_3^0 Z) \\ &\sim (a_2^2 Y, a_2^2 Z) \\ &\sim (Y, Z)\end{aligned}$$

On the other hand, if $a_2^1 < a_1^1$, then there exists $\theta \in \mathbb{R}$ such that $a_1^1 \sinh \theta + a_2^1 \cosh \theta = 0$. Hence

$$\begin{aligned}(A, B) &\sim (a_1^2 X, b_3^0 Z) \\ &\sim (a_1^2 X, a_1^2 Z) \\ &\sim (X, Z)\end{aligned}$$

2.6 Summary

Proposition 2.2. Control systems of the form (3) on each 3D unimodular Lie group are equivalent to one of the model systems for the 3D unimodular Lie group, as listed in Table 1.

3 Small-Time Controllability

Assume the equivalent system is (f_0, f_1) , or

$$\dot{g} = (f_0 + u f_1)g \tag{4}$$

with $g \in G$ and $f_0, f_1 \in \mathfrak{g}$.

Lemma 3.1. If the pair (f_0, f_1) is not bracket generating, then the system 4 is not small-time controllable.

Lie group	Model systems
$SU(2)$	(σ_1, σ_3)
$SL(2)$	(E, H) (F, H) $(E + F, H)$ $(E, E - F)$ (H, E) (F, E)
H	(X, Z) (Z, X) (Y, X)
$SE(2)$	(Y, X) (Z, X) (X, Z)
$SH(2)$	(X, Z) (Y, Z) $(X + Y, Z)$ $(X, X + Y)$ $(Z, X + Y)$ (Z, Y) (X, Y) (Z, X) (Y, X)

Table 1: Summary of Model Control Systems

3.1 Controllability on $SU(2)$

First,

$$[u_1, u_3] = \begin{pmatrix} 0 & -1/2 \\ 1/2 & 0 \end{pmatrix} = -\frac{1}{2}u_2$$

so

$$\text{span}\{u_1, u_3, [u_1, u_3]\} = \text{span}\{u_1, u_3, -\frac{1}{2}u_2\} = \mathfrak{su}(2)$$

and so the pair is bracket generating. We may proceed to the next step.

$$e^{vu_3} = e^{\begin{pmatrix} 0 & i/2 \\ i/2 & 0 \end{pmatrix}} = \begin{pmatrix} \cos \frac{v}{2} & i \sin \frac{v}{2} \\ i \sin \frac{v}{2} & \cos \frac{v}{2} \end{pmatrix}$$

In this case, the closure $\overline{e^{vu_3}} = e^{vu_3}$, and we will denote this set L . Then,

$$e^{vu_3}u_1e^{-vu_3} = \begin{pmatrix} \frac{i}{2}\cos t & \frac{1}{2}\sin t \\ -\frac{1}{2}\sin t & -\frac{i}{2}\cos t \end{pmatrix} = \cos(t)u_1 + \sin(t)u_2$$

hence

$$Ad_L u_1 = \{\cos(t)u_1 + \sin(t)u_2 : t \in \mathbb{R}\}$$

Furthermore,

$$Ad_L(u_1 + uu_3) = \{\cos(t)u_1 + \sin(t)u_2 + uu_3 : t \in \mathbb{R}\}$$

This represents a cylinder in the u_1 - u_2 - u_3 space, whose axis passes through the origin and is perpendicular to the u_1 - u_2 plane. We then convexify this set, and quotient out the “fast” direction u_3 .

$$\text{conv}(Ad_L u_1)/\mathbb{R}u_3 = \text{conv}\{\cos(t)u_1 + \sin(t)u_2 : t \in \mathbb{R}\}$$

This is a bounded disc in the u_1 - u_2 plane. Since the convexified set is bounded, the system (u_1, u_3) is not small-time controllable.

Remark. It should not matter whether we convexify first or quotient first.

3.2 Controllability on $SL(2)$

3.2.1 (E, H)

Notice that $[E, H] = -2E$, and so the F direction does not appear. The system is not bracket generating, and so it cannot be small-time controllable. Another way to see this is that linear combinations of E and H generate upper-triangular matrices of the form

$$\begin{pmatrix} b & a \\ 0 & -b \end{pmatrix}$$

The constant 0 makes it clear that the system cannot be small-time controllable.

3.2.2 (F, H)

This case is similar to the previous one. The commutator $[F, H] = 2F$, and so the E direction does not appear. The system is not bracket generating, thus it cannot be small-time controllable. Another way to see this is that linear combinations of F and H generate lower-triangular matrices, from which it is obvious that the system cannot be small-time controllable.

3.2.3 $(E + F, H)$

The convexified set, after we quotient out the “fast” direction H , is the region above one branch of a hyperbola in the E - F plane, given by

$$\text{conv}(Ad_{e^{vH}}(E + F))/\mathbb{R}H = \text{conv}\{e^{2v}E + e^{-2v}F : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vH}}(E + F)) + uH = \text{conv}\{e^{2v}E + e^{-2v}F : v \in \mathbb{R}\} + uH$$

Moving to the limit, we can achieve the positive half of an infinite line in the direction of E , F , or some linear combination of E and F . For E and F , we have

$$[E, H] = -2E \in \text{span}\{E, H\}, \quad [F, H] = 2F \in \text{span}\{F, H\}$$

On the other hand, for a linear combination $\alpha E + \beta F$,

$$[\alpha E + \beta F, H] = -2\alpha E + 2\beta F \notin \text{span}\{\alpha E + \beta F, H\}$$

meaning this choice generates a new direction. With that being said, the fact that we only have a half-line poses a major obstacle in either case.

If we would like to continue to quotient out both H and the new direction, the main problem is that we only have the positive half of the line. We cannot directly employ the inverse of the matrix exponential, for example e^{-uE} , when computing $Ad_{e^{uE+vH}}(E + F)$. Moreover, we also cannot do

$$SL(2)/e^{uE+vH}$$

to check periodicity, because under the restriction $u > 0$, e^{uE+vH} cannot form a subgroup.

We employ the barrier function method instead. The control vector flow is

$$\exp(uH) = \begin{pmatrix} e^u & 0 \\ 0 & e^{-u} \end{pmatrix}$$

For

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2) \text{ with } ad - bc = 1,$$

$$\exp(uH)g = \begin{pmatrix} ae^u & be^u \\ ce^{-u} & de^{-u} \end{pmatrix}$$

Conceptually, we want to construct a function F that is invariant under the control. So this serves as motivation to choose

$$F(g(t)) = ac + bd$$

Now we transform the matrix representation into a state-space representation. From

$$\dot{g} = (E + F + uH)g$$

we have

$$\begin{pmatrix} \dot{a} & \dot{b} \\ \dot{c} & \dot{d} \end{pmatrix} = \begin{pmatrix} au + c & bu + d \\ -cu + a & -du + b \end{pmatrix}$$

and so

$$\begin{cases} \dot{a} &= au + c \\ \dot{b} &= bu + d \\ \dot{c} &= -cu + a \\ \dot{d} &= -du + b \end{cases}$$

We might guess from this structure that taking $F(g(t))$ as above could yield a derivative that is useful to us. Indeed,

$$\begin{aligned} \dot{F}(g(t)) &= \dot{a}c + a\dot{c} + \dot{b}d + b\dot{d} \\ &= (au + c)c + a(-cu + a) + (bu + d)d + b(-du + b) \\ &= a^2 + b^2 + c^2 + d^2 \\ &\geq 0 \end{aligned}$$

Assume $a^2 + b^2 + c^2 + d^2 = 0$. Since $a, b, c, d \geq 0$, this implies $a = b = c = d = 0$. This contradicts the restriction $ad - bc = 1$. Hence, in fact,

$$\dot{F}(g(t)) > 0$$

for all trajectories $g(t)$ of the system $(E + F, H)$. We can also check that at the identity,

$$F(I) = 1 \cdot 0 + 0 \cdot 1 = 0$$

So $F > 0$ for any trajectory of the system. Lastly, consider

$$\exp(-\varepsilon F) = \exp\left(\begin{pmatrix} 0 & 0 \\ -\varepsilon & 0 \end{pmatrix}\right) = \begin{pmatrix} 1 & 0 \\ -\varepsilon & 1 \end{pmatrix},$$

arbitrarily close to the identity. We have

$$F(\exp(-\varepsilon F)) = -\varepsilon < 0, \forall \varepsilon > 0$$

This is a point that cannot be reached by the trajectories in positive time. This proves $(E+F, H)$ is not small-time locally controllable, hence not small-time controllable.

Alternatively, we can choose

$$\exp(\varepsilon H - \varepsilon E) = \begin{pmatrix} e^\varepsilon & -\sinh \varepsilon \\ 0 & e^{-\varepsilon} \end{pmatrix}$$

close to the identity for small $\varepsilon > 0$.

The same method applies to the system $(-E - F, H)$. By choosing

$$F(g(t)) = ac + bd,$$

we have

$$\begin{aligned} F(I) &= 0 \\ \dot{F}(g(t)) &= -a^2 - b^2 - c^2 - d^2 < 0. \end{aligned}$$

We can then choose

$$F(\exp(\varepsilon F)) = \varepsilon > 0,$$

proving that $(-E - F, H)$ is also not small-time controllable.

3.2.4 $(E, E - F)$

Convexified set is the region bounded by an ellipse. Hence the system is not small-time controllable.

3.2.5 (H, E)

Not bracket generating, hence not small-time controllable.

3.2.6 (F, E)

The convexified set, after we quotient out the “fast” direction E , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vE}}F)/\mathbb{R}E = \{F + vH : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vE}}F) + uE = \{F + uE + vH : u, v \in \mathbb{R}\}$$

Here, we have unlocked a new direction for the control, H . Now, the E and H directions can be made arbitrarily large by taking large u and v . This effectively makes F negligible. We check the bracket

$$[H, E] = 2E \in \text{span}\{H, E\}$$

E and H were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (F + uE + vH)g$$

By taking the commutator, we know that the (closure of the) reachable set is the same as the original system.

Remark. Ideally, we want $[H, E]$ to not be in the span of H and E , as that would give us a new direction. When it is in the span of H and E , the new system we are considering above is still valid. It is because no new direction was generated that we need to continue to quotient (otherwise, with two fast directions generating the third direction, we would be done).

This new system has two fast directions, E and H , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uE+vH}}F)/\text{span}\{E, H\} = \{e^{-2v}F : v \in \mathbb{R}\}$$

This one-dimensional set is the positive half of an infinite line in the F direction, not including the origin. It is appropriate to take the closure of the set to include the origin?

Remark. When computing $Ad_{e^{uE+vH}}F$, we need the inverse matrix exponential, e^{-uE-vH} . Mathematically, this is possible for any matrix. However, possible restrictions on the controls mean we need to be more cautious. After the first quotient, the convexified set was an infinite straight line not passing through the origin in the F - H plane. By taking points on the line to the limit (taking large v), we can approximate an infinite line along the H axis passing the origin, that most importantly, goes in both directions. We were able to obtain both directions for free. Thanks to this, we can indeed directly use e^{-uE-vH} in this scenario. This is different from the hyperbola cases, where we only have the positive half of the line.

Let us now check if the solution to the system

$$\dot{g} = (wF + uE + vH)g, \quad w > 0$$

is periodic. If it is, then the periodicity allows this set to be viewed as an infinite line passing through the origin. Suppose we take $v = 0$, $u = -1$, $w = 1$ so that

$$wF + uE + vH = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

The solution to this matrix differential equation is

$$\begin{aligned} g(t) &= \exp \left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} t \right) g(0) \\ &= [\cos(t)I + \sin(t)(F - E)]g(0) \end{aligned}$$

which is a periodic solution. In fact, the solution will always be periodic given $v = 0$, $u = -w$ with $w > 0$. This suggests that the set we found can be seen as an infinite line through the origin.

Moreover, we can examine

$$SL(2)/e^{uE+vH}$$

where e^{uE+vH} is interpreted as a subgroup of $SL(2)$. The result must be a one-dimensional manifold, which is completely classified as a line $\mathbb{R}$ and a circle S^1 . Using the KAN decomposition, we have

$$SL(2)/e^{uE+vH} \cong S^1$$

This result supports the periodic solution we just found.

3.3 Controllability on H

3.3.1 (X, Z)

Not bracket generating, hence not small-time controllable.

3.3.2 (Z, X)

Not bracket generating, hence not small-time controllable.

3.3.3 (Y, X)

The convexified set, after we quotient out the “fast” direction X , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vX}}Y)/\mathbb{R}X = \{Y + vZ : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vX}}Y) + uX = \{Y + uX + vZ : u, v \in \mathbb{R}\}$$

Now, the X and Z directions can be made arbitrarily large by taking large u and v . This effectively makes Y negligible. We check the bracket

$$[Z, X] = 0 \in \text{span}\{Z, X\}$$

X and Z were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (Y + uX + vZ)g$$

By taking the commutator, we know that the (closure of the) reachable set is the same as the original system. This new system has two fast directions, X and Z , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uX+vZ}}Y)/\text{span}\{X, Z\}$$

We compute

$$e^{uX+vZ} = \exp \left(\begin{pmatrix} 0 & u & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 & u & v \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : u, v \in \mathbb{R} \right\}$$

So

$$Ad_{e^{uX+vZ}}Y = \begin{pmatrix} 1 & 0 & u \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \{Y + uZ : u \in \mathbb{R}\}$$

Convexifying this set,

$$\text{conv}(Ad_{e^{uX+vZ}}Y) = \{Y + uZ : u \in \mathbb{R}\}$$

We then quotient out the “fast” X and Z directions, giving

$$\text{conv}(Ad_{e^{uX+vZ}}Y)/\text{span}\{X, Z\} = \{Y\}$$

which is a singleton, a one-dimensional bounded set. This suggests the system is not small-time controllable.

3.4 Controllability on $SE(2)$

3.4.1 (Y, X)

Not bracket generating, hence not small-time controllable.

3.4.2 (Z, X)

The convexified set, after we quotient out the “fast” direction X , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vX}}Z)/\mathbb{R}X = \{Z - vY : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vX}}Z) + uX = \{Z + uX - vY : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily large by taking large u and v . This effectively makes Z negligible. We check the bracket

$$[Y, X] = 0 \in \text{span}\{Y, X\}$$

X and Y were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (Z + uX - vY)g$$

By taking the commutator, we know that the (closure of the) reachable set is the same as the original system. This new system has two fast directions, X and Y , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uX-vY}}Z)/\text{span}\{X, Y\}$$

We compute

$$e^{uX-vY} = \exp \left(\begin{pmatrix} 0 & 0 & u \\ 0 & 0 & -v \\ 0 & 0 & 0 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 & 0 & u \\ 0 & 1 & -v \\ 0 & 0 & 1 \end{pmatrix} : u, v \in \mathbb{R} \right\}$$

So

$$Ad_{e^{uX-vY}}Z = \begin{pmatrix} 0 & -1 & -v \\ 1 & 0 & -u \\ 0 & 0 & 0 \end{pmatrix} = \{Z - vX - uY : u, v \in \mathbb{R}\}$$

Convexifying this set gives a plane passing through Z parallel to the X - Y plane. We then quotient out the “fast” X and Y directions, giving

$$\text{conv}(Ad_{e^{uX-vY}}Z)/\text{span}\{X, Y\} = \{Z\}$$

which is a singleton, a one-dimensional bounded set. This suggests the system is not small-time controllable.

3.4.3 (X, Z)

Convexified set is the region bounded by a circle. Hence the system is not small-time controllable.

3.5 Controllability on $SH(2)$

3.5.1 (X, Z)

The convexified set, after we quotient out the “fast” direction Z , is the region to the right of one branch of a hyperbola in the X - Y plane, given by

$$\text{conv}(Ad_{e^{vZ}}X)/\mathbb{R}Z = \text{conv}\{\cosh vX + \sinh vY : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vZ}}X) + uZ = \text{conv}\{\cosh vX + \sinh vY : v \in \mathbb{R}\} + uZ$$

Moving to the limit, we can achieve the positive half of an infinite line in the direction of $X + Y$ or $X - Y$ (along the asymptotes of the hyperbola), or some other linear combination of X and Y that keeps the line within the convexified set. For $X + Y$ and $X - Y$, we have

$$[X + Y, Z] = -Y - X \in \text{span}\{X + Y, Z\}, \quad [X - Y, Z] = -Y + X \in \text{span}\{-X + Y, Z\}.$$

Another obvious choice of direction would be X , with the distinction that

$$[X, Z] = -Y \notin \text{span}\{X, Z\},$$

Or more generally, taking $\alpha > 0, \beta > 0$,

$$[\alpha X + \beta Y, Z] = -\alpha Y - \beta X \notin \text{span}\{\alpha X + \beta Y, Z\}$$

unless $\alpha = \beta$, and

$$[\alpha X - \beta Y, Z] = -\alpha Y + \beta X \notin \text{span}\{\alpha X - \beta Y, Z\}$$

unless $\alpha = \beta$. So these choices are able to generate new directions.

Similar to the $(E + F, H)$ case in $SL(2)$, we cannot directly employ the inverse of the matrix exponential when computing the conjugate. We try to apply the barrier function method again.

$SH(2)$ can be written as a (inner) semidirect product

$$SH(2) = N \rtimes H$$

where

$$N = \left\{ \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} : x, y \in \mathbb{R} \right\} = \left\{ \begin{pmatrix} I & v \\ 0 & 1 \end{pmatrix} : v \in \mathbb{R}^2 \right\}$$

$$H = \left\{ \begin{pmatrix} \cosh \theta & \sinh \theta & 0 \\ \sinh \theta & \cosh \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} : \theta \in \mathbb{R} \right\} = \left\{ \begin{pmatrix} R(\theta) & 0 \\ 0 & 1 \end{pmatrix} : \theta \in \mathbb{R} \right\}$$

Indeed, N is a normal subgroup of G , H is a subgroup of G , $SH(2) = NH$, and $N \cap H = \{I\}$.

Again, we would like a F that is invariant under the control. We may consider the concepts of spatial coordinates and body coordinates from rigid body mechanics. The body coordinates remain invariant when the object is in motion. Let x denote the spatial coordinates and let X denote the body coordinates, then we can transform between the two by

$$x = R(\theta)X$$

$$X = R(\theta)^{-1}x$$

In this case,

$$R(\theta)^{-1} = \begin{pmatrix} \cosh \theta & -\sinh \theta \\ -\sinh \theta & \cosh \theta \end{pmatrix} = R(-\theta)$$

Let us transform the spatial coordinates (x, y) into the body coordinates (ξ, η) .

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = R(\theta)^{-1} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \cosh \theta - y \sinh \theta \\ y \cosh \theta - x \sinh \theta \end{pmatrix}$$

Remark. We can also work with (x, y, θ) and (ξ, η, θ) , taking

$$R(\theta) = \begin{pmatrix} \cosh \theta & \sinh \theta & 0 \\ \sinh \theta & \cosh \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

which gives the same result.

This is some motivation for choosing

$$F(g(t)) = \xi = x \cosh \theta - y \sinh \theta$$

Now, let us transform the matrix representation of the system (X, Z) into the state-space representation. From

$$\dot{g} = (X + uZ)g$$

we have

$$\begin{pmatrix} \sinh \theta \dot{\theta} & \cosh \theta \dot{\theta} & \dot{x} \\ \cosh \theta \dot{\theta} & \sinh \theta \dot{\theta} & \dot{y} \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} u \sinh \theta & u \cosh \theta & uy + 1 \\ u \cosh \theta & u \sinh \theta & ux \\ 0 & 0 & 0 \end{pmatrix}$$

so

$$\begin{cases} \dot{x} &= uy + 1 \\ \dot{y} &= ux \\ \dot{\theta} &= u \end{cases}$$

Using the above choice of $F(g(t))$

$$F(I) = 0$$

$$\begin{aligned} \dot{F}(g(t)) &= \dot{x} \cosh \theta + x \sinh \theta \dot{\theta} - \dot{y} \sinh \theta - y \cosh \theta \dot{\theta} \\ &= (uy + 1) \cosh \theta + ux \sinh \theta - ux \sinh \theta - uy \cosh \theta \\ &= \cosh \theta \\ &\geq 1 \\ &> 0 \end{aligned}$$

At a point close to the identity,

$$F(\exp(-\varepsilon X)) = F\left(\begin{pmatrix} 1 & 0 & -\varepsilon \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\right) = -\varepsilon < 0, \quad \forall \varepsilon > 0$$

This proves that (X, Z) is not small-time locally controllable, hence not small-time controllable.

Remark. We can also work with vector fields. For a system

$$\dot{g} = (f_0 + \sum_{i=1}^n u_i f_i)g,$$

we have

$$\frac{d}{dt}F(g(t)) = V_{f_0}F(g(t)) = \sum_{i=1}^n u_i V_{f_i}F(g(t))$$

In this case,

$$\dot{F}(g(t)) = V_X F(g(t)) + u V_Z F(g(t)).$$

Moreover,

$$V_Z \xi = V_Z \eta = 0$$

ξ and η are first integrals of the control vector field.

Remark. Choosing $F(g(t)) = \eta = y \cosh \theta - x \sinh \theta$ would not work because $\dot{F}(g(t)) = -\sinh \theta$ is not greater than zero.

3.5.2 (Y, Z)

The convexified set, after we quotient out the “fast” direction Z , is the region above one branch of a hyperbola in the X - Y plane, given by

$$\text{conv}(Ad_{e^{vZ}}Y)/\mathbb{R}Z = \text{conv}\{\sinh vX + \cosh vY : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vZ}}Y) + uZ = \text{conv}\{\sinh vX + \cosh vY : v \in \mathbb{R}\} + uZ$$

Moving to the limit, we can achieve the positive half of an infinite line in the direction of $X + Y$ or $-X + Y$ (along the asymptotes of the hyperbola), or some other linear combination of X and Y that keeps the line within the convexified set. For $X + Y$ and $-X + Y$, we have

$$[X + Y, Z] = -Y - X \in \text{span}\{X + Y, Z\}, \quad [-X + Y, Z] = Y - X \in \text{span}\{-X + Y, Z\}.$$

Another obvious choice of direction would be Y , with the distinction that

$$[Y, Z] = -X \notin \text{span}\{Y, Z\},$$

Or more generally, taking $\alpha > 0$, $\beta > 0$,

$$[\alpha X + \beta Y, Z] = -\alpha Y - \beta X \notin \text{span}\{\alpha X + \beta Y, Z\}$$

unless $\alpha = \beta$, and

$$[-\alpha X + \beta Y, Z] = \alpha Y - \beta X \notin \text{span}\{-\alpha X + \beta Y, Z\}$$

unless $\alpha = \beta$. So these choices are able to generate new directions.

Similar to the $(E + F, H)$ case in $SL(2)$, we cannot directly employ the inverse of the matrix exponential when computing the conjugate. The method we will follow is almost identical to the (X, Z) case.

First, we write the matrix differential equation

$$\dot{g} = (Y + uZ)g$$

in its state-space representation.

$$\begin{pmatrix} \sinh \theta \dot{\theta} & \cosh \theta \dot{\theta} & \dot{x} \\ \cosh \theta \dot{\theta} & \sinh \theta \dot{\theta} & \dot{y} \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} u \sinh \theta & u \cosh \theta & uy \\ u \cosh \theta & u \sinh \theta & ux + 1 \\ 0 & 0 & 0 \end{pmatrix}$$

and so

$$\begin{cases} \dot{x} &= uy \\ \dot{y} &= ux + 1 \\ \dot{\theta} &= u \end{cases}$$

We choose

$$F(g(t)) = \eta = y \cosh \theta - x \sinh \theta$$

which gives

$$F(I) = 0$$

$$\begin{aligned} \dot{F}(g(t)) &= \dot{y} \cosh \theta + y \sinh \theta \dot{\theta} - \dot{x} \sinh \theta - x \cosh \theta \dot{\theta} \\ &= (ux + 1) \cosh \theta + uy \sinh \theta - uy \sinh \theta - ux \cosh \theta \\ &= \cosh \theta \\ &\geq 1 \\ &> 0 \end{aligned}$$

and at a point near the identity,

$$F(\exp(-\varepsilon Y)) = F\left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -\varepsilon \\ 0 & 0 & 1 \end{pmatrix}\right) = -\varepsilon < 0, \quad \forall \varepsilon > 0$$

This proves that the system (Y, Z) is not small-time locally controllable, and therefore not small-time controllable.

3.5.3 $(X + Y, Z)$

Not bracket generating, hence not small-time controllable.

3.5.4 $(X, X + Y)$

Not bracket generating, hence not small-time controllable.

3.5.5 $(Z, X + Y)$

Not bracket generating, hence not small-time controllable.

3.5.6 (Z, Y)

The convexified set, after we quotient out the “fast” direction Y , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vY}}Z)/\mathbb{R}Y = \{Z - vX : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vY}}Z) + uY = \{Z + uY - vX : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily large by taking large u and v . This effectively makes Z negligible. We check the bracket

$$[X, Y] = 0 \in \text{span}\{X, Y\}$$

X and Y were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (Z + uY - vX)g$$

By taking the commutator, we know that the (closure of the) reachable set is the same as the original system. This new system has two fast directions, X and Y , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uY-vX}}Z)/\text{span}\{X, Y\}$$

We compute

$$e^{uY-vX} = \exp \left(\begin{pmatrix} 0 & 0 & -v \\ 0 & 0 & u \\ 0 & 0 & 0 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 & 0 & -v \\ 0 & 1 & u \\ 0 & 0 & 1 \end{pmatrix} : u, v \in \mathbb{R} \right\}$$

So

$$Ad_{e^{uY-vX}}Z = \begin{pmatrix} 0 & 1 & -u \\ 1 & 0 & v \\ 0 & 0 & 0 \end{pmatrix} = \{Z - uX + vY : u, v \in \mathbb{R}\}$$

Convexifying this set gives a plane passing through Z parallel to the X - Y plane. We then quotient out the “fast” X and Y directions, giving

$$\text{conv}(Ad_{e^{uY-vX}}Z)/\text{span}\{X, Y\} = \{Z\}$$

which is a singleton, a one-dimensional bounded set. This suggests the system is not small-time controllable.

3.5.7 (X, Y)

Not bracket generating, hence not small-time controllable.

3.5.8 (Z, X)

The convexified set, after we quotient out the “fast” direction X , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vX}}Z)/\mathbb{R}X = \{Z - vY : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vX}}Z) + uX = \{Z + uX - vY : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily large by taking large u and v . This effectively makes Z negligible. We check the bracket

$$[Y, X] = 0 \in \text{span}\{Y, X\}$$

X and Y were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (Z + uX - vY)g$$

By taking the commutator, we know that the (closure of the) reachable set is the same as the original system. This new system has two fast directions, X and Y , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uX-vY}}Z)/\text{span}\{X, Y\}$$

We compute

$$e^{uX-vY} = \exp \left(\begin{pmatrix} 0 & 0 & u \\ 0 & 0 & -v \\ 0 & 0 & 0 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 & 0 & u \\ 0 & 1 & -v \\ 0 & 0 & 1 \end{pmatrix} : u, v \in \mathbb{R} \right\}$$

So

$$Ad_{e^{uX-vY}}Z = \begin{pmatrix} 0 & 1 & v \\ 1 & 0 & -u \\ 0 & 0 & 0 \end{pmatrix} = \{Z + vX - uY : u, v \in \mathbb{R}\}$$

Convexifying this set gives a plane passing through Z parallel to the X - Y plane. We then quotient out the “fast” X and Y directions, giving

$$\text{conv}(Ad_{e^{uX-vY}}Z)/\text{span}\{X, Y\} = \{Z\}$$

which is a singleton, a one-dimensional bounded set. This suggests the system is not small-time controllable.

3.5.9 (Y, X)

Not bracket generating, hence not small-time controllable.

3.6 Summary

The following table summarizes whether each of the model systems found is small-time controllable or not. Among all the model systems found, the only system that is small-time controllable is (F, E) .

$SU(2)$	(u_1, u_3)	No
$SL(2)$	(E, H)	No
	(F, H)	No
	$(\pm E \pm F, H)$	No
	$(E, E - F)$	No
	(H, E)	No
	(F, E)	Yes
H	(X, Z)	No
	(Z, X)	No
	(Y, X)	No
$SE(2)$	(Y, X)	No
	(Z, X)	No
	(X, Z)	No
$SH(2)$	(X, Z)	No
	(Y, Z)	No
	$(X + Y, Z)$	No
	$(X, X + Y)$	No
	$(Z, X + Y)$	No
	(Z, Y)	No
	(X, Y)	No
	(Z, X)	No
	(Y, X)	No

Table 2: Summary of Small-Time Controllability

4 Conclusion

Throughout this paper, we first classified all control systems of the form 3 on all 3D unimodular matrix Lie groups, and then determined their small-time controllability. 22 equivalent systems were found in total, of which only one, the system (F, E) on $SL(2)$, was small-time controllable.

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